Assignment 1

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Question 1

(a) They will be able to see sunrise from the **vernal equinox** to the **autumnal equinox**.
Reason: People will be able to see the sunrise if the azimuth is less than 90 degrees when the sun rises. The sun rises from the north-east during this period, so the azimuth is less than 90 degrees.

(b) They will be able to glimpse the sun at noon around **June**.
Reason: People will be able to glimpse the sun at noon if the zenith cross happens. Hong Kong's latitude is 22.5 degrees, which means the zenith cross happens near June.

Question 2

(a) The ones on the **south** side of the street.
Reason: Tokyo's latitude is 35.5 degrees. So, during the whole year, the sun is at the south of zenith at noon. This will lead to the shadows pointing toward the north.

(b) The shadows are longest at the **winter solstice**.
Reason: During the winter solstice, the altitude of the sun is the lowest, which means that the shadows are the longest.

Question 3

She faces **north**.
Reason: As a northern observer, all the stars rotate around the north celestial pole, which is tilted towards the north. Therefore, the stars near the north celestial pole will never set below the horizon when facing north.

Question 4

(a) Hong Kong has a **higher** noon solar altitude than Beijing on **both the autumnal equinox** and **the winter solstice**.
Reason: Hong Kong's latitude is 22.5 degrees, while Beijing is 40 degrees. On the autumnal equinox, the altitude of the sun is 90 - latitude. On winter solstice, the altitude of the sun is 90 - 23.5 - latitude.

(b) Compared to Beijing, the azimuth of the sunrise is **lower** in Hong Kong on the **winter solstice**, but the **same** on the **vernal equinox**.
Reason: For the northern hemisphere, the sunrise azimuth is at the southeast in winter solstice. The higher the altitude, the more the sunrise azimuth shifts south. However, on the vernal equinox, the sun rises on the east regardless of the altitude.

(c) The length of the daylight is the **same** on the **vernal equinox**. On the **summer solstice**, compared to Beijing, the length of the daylight is **shorter** in Hong Kong.
Reason: On the vernal equinox, the daylights are the same everywhere (12h). On the summer solstice, the length of daylights are longer than those of the nights. The higher the altitude, the longer the length of daylight (by observing the orbits of the sun above the horizon).

Question 5

Evidence 1: The shadow on the moon is round.
Reason: The shadow on the moon is casted by Earth. Since the shadow is round, the Earth must be round.

Evidence 2: By analogy with the fact that the Moon and the Sun are spherical.
Reason: When ancient astronomers observe the moon and the sun, they may compare their shape with Earth, which may lead them to conclude that Earth is round.